



Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

- VERIFIED 0.80 - 1.00** 3+ independent sources including media
- CONFIRMED 0.50 - 0.79** 2 sources or 1 high-quality media
- EMERGING 0.25 - 0.49** Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	183	Media
TechCrunch AI	10	Media
HackerNews	6	Community
The Verge AI	5	Media
VentureBeat AI	1	Media
MIT Tech Review	1	Media

VERIFIED INTELLIGENCE — 3 stories

VERI

OpenClaw and Claude can put your AI-generated podcasts in Spotify

The Verge AI

+1 source

80%

Save to Spotify is a new command-line tool designed specifically for AI agents like OpenClaw, Claude Code, or OpenAI Codex. If you're the kind of person who collects research on a topic, then feeds it through their AI of choice to create audio summaries and personal podcasts,...

<https://www.theverge.com/entertainment/925916/save-to-spotify-ai-podcasts>

VERI

SpaceX has a \$55 billion plan to build AI chips in Texas

The Verge AI

+1 source

80%

Elon Musk's plans to get into the AI chip manufacturing business are going to be costly. As the New York Times and CNBC report, SpaceX is planning to invest at least \$55 billion into its "Terafab" chip plant in Austin, Texas. That's according to the details of a public hearing...

<https://www.theverge.com/ai-artificial-intelligence/926356/spacex-terafab-plant-cost-ai...>

VERI

DataDignity: Training Data Attribution for Large Language Models

ArXiv CS.AI

+1 source

80%

arXiv:2605.05687v1 Announce Type: new Abstract: Auditing language-model outputs often requires more than judging correctness: an auditor may need to identify which source document most likely supports the knowledge expressed in a response. We study this as pinpoint provenance:...

<https://arxiv.org/abs/2605.05687>

CONFIRMED SIGNALS — 200 stories

CONF

Show HN: Stage CLI - An easier way of reading your AI generated changes locally

HackerNews

50%

HN 35

Hi All, Recently I've been using Claude Code a lot for debugging cluster issues and I realized I was performing similar tasks repeatedly so I decided to package them up into skills so I could call them up more easily (e.g. ``/investigate``, ``/audit-security``, ``/audit-outdated``)....

<https://github.com/ReviewStage/stage-cli>

CONF

Show HN: I made a vertical-peddalling bike with a novel drivetrain [video]

HackerNews

50%

HN 21

Hi HN! I made an app that takes a photo of a paper drawing and, in a handful of seconds, creates a fully rigged character that can be used in an animation or little story. It doesn't use any image-to-video generative AI models. Instead, I built it using the years of insights...

<https://www.youtube.com/watch?v=4HLOsi2gWXQ>

CONF

Show HN: Platos - like Claude Managed Agents but open-source and self-hosted

HackerNews

50%

HN 8

No summary — see link for full article.

<https://github.com/winsenlabs/platos>

CONF

Show HN: Local-first long-term memory engine for AI agents·MCP/CLI· 100% local

HackerNews 50% HN 4

Local-first long-term memory engine for AI agents · MCP + HTTP + CLI · SQLite + sqlite-vec + FTS5 · 100% local, no cloud
<https://github.com/misaelzapata/memoirs>

CONF

Show HN: AICW Video open-source to cut video into clips with captions, voiceover

HackerNews 50% HN 4

No summary — see link for full article.

<https://github.com/aicw-io/aicw-video>

CONF

Show HN: AnamDB - An AI-native, differentiable Datalog engine written in Rust

HackerNews 50% HN 2

No summary — see link for full article.

<https://github.com/jam5991/anam>

0 CONF

Voi founders' new AI startup Pit has become the latest rising star out of Stockholm

TechCrunch AI 50%

AI startup Pit is led by the co-founders of European scooter giant Voi and backed by a16z, which is leading the startup's \$16 million seed round.

<https://techcrunch.com/2026/05/07/voi-founders-new-ai-startup-pit-has-become-the-latest...>

1 CONF

OpenAI introduces new 'Trusted Contact' safeguard for cases of possible self-harm

TechCrunch AI 50%

The company is expanding its efforts to protect ChatGPT users in cases where conversations may turn to self-harm.

<https://techcrunch.com/2026/05/07/openai-introduces-new-trusted-contact-safeguard-for-c...>

2 CONF

Perplexity's Personal Computer is now available to everyone on Mac

TechCrunch AI 50%

Perplexity's Personal Computer brings AI agents to your Mac, and is now open to everyone.

<https://techcrunch.com/2026/05/07/perplexitys-personal-computer-is-now-available-everyo...>

3 CONF

Elon Musk's lawsuit is putting OpenAI's safety record under the microscope

TechCrunch AI 50%

Elon Musk's legal effort to dismantle OpenAI may hinge on how its for-profit subsidiary enhances or detracts from the frontier lab's founding mission of ensuring that humanity benefits from artificial general intelligence.

<https://techcrunch.com/2026/05/07/elon-musks-lawsuit-is-putting-openais-safety-record-u...>

4 CONF

How Anthropic's Mythos has rewritten Firefox's approach to cybersecurity

TechCrunch AI 50%

Security researchers at Mozilla say Anthropic's Mythos has unearthed a wealth of high-severity bugs in Firefox.

<https://techcrunch.com/2026/05/07/how-anthropics-mythos-has-rewritten-firefoxs-approach...>

5 CONF

Startup Battlefield 200 applications close May 27: A shot at VC access, global visibility, TechCrunch coverage, and \$100K

TechCrunch AI 50%

Startup Battlefield 200 applications are open, but only for three more weeks. Apply by May 27 for your shot at VC access, global visibility, TechCrunch coverage, \$100,000 equity-free, and more opportunities for major scaling impact.

<https://techcrunch.com/2026/05/07/startup-battlefield-200-applications-close-may-27-a-s...>

6 CONF

China's Moonshot AI raises \$2B at \$20B valuation as demand for open source AI skyrockets

TechCrunch AI 50%

Moonshot's annualized recurring revenue topped \$200 million in April, driven by rapid growth in paid subscriptions and API usage.

<https://techcrunch.com/2026/05/07/chinas-moonshot-ai-raises-2b-at-20b-valuation-as-dema...>

7 CONF

Spotify wants to become the home for AI-generated personal audio

TechCrunch AI 50%

Users will be able to create a podcast from Codex or Claude Code and import it to Spotify.

<https://techcrunch.com/2026/05/07/spotify-wants-to-become-the-home-for-ai-generated-per...>

8 CONF

Spotify's AI DJ now supports French, German, Italian, and Brazilian Portuguese

TechCrunch AI 50%

Spotify's AI DJ feature now supports French, German, Italian, and Brazilian Portuguese.

<https://techcrunch.com/2026/05/07/spotify-s-ai-dj-now-supports-french-german-italian-and...>

9 CONF

Apple's AirPods with cameras for AI are apparently close to production

The Verge AI 50%

Apple's rumored AirPods with cameras are nearing a stage where the company will test early mass production, Bloomberg's Mark Gurman reports. Currently, Apple testers are "actively using" prototypes that are in the design validation test stage, which is one step before the...

<https://www.theverge.com/tech/926376/apple-airpods-cameras-ai-production>

10 CONF

ChatGPT's Trusted Contact will alert loved ones of safety concerns

The Verge AI 50%

OpenAI is launching an optional safety feature for ChatGPT that allows adult users to assign an emergency contact for mental health and safety concerns. Friends, family members, or caregivers designated as a "Trusted Contact" will be notified if OpenAI detects that a person may...

<https://www.theverge.com/ai-artificial-intelligence/925874/chatgpt-trusted-contact-emer...>

1 CONF

Google's taking a big swing at AI health with the Fitbit Air

The Verge AI 50%

It's a Whoop dupe. That was my first thought when I saw the new \$99 Google Fitbit Air. You can hardly blame me. The band is screenless with a metallic fabric clasp. My eyes flickered between the Fitbit Air and my wrist, where I'm wearing a Whoop MG. Was I not seeing double? But...

<https://www.theverge.com/gadgets/925458/google-health-fitbit-air-ai-coaching-wearables-...>

2 CONF

Partial Evidence Bench: Benchmarking Authorization-Limited Evidence in Agentic Systems

ArXiv CS.AI 50%

arXiv:2605.05379v1 Announce Type: new Abstract: Enterprise agents increasingly operate inside scoped retrieval systems, delegated workflows, and policy-constrained evidence environments. In these settings, access control can be enforced correctly while the system still produces...

<https://arxiv.org/abs/2605.05379>

3 CONF

Intelligent CCTV for Urban Design: AI-Based Analysis of Soft Infrastructure at Intersections

ArXiv CS.AI 50%

arXiv:2605.05402v1 Announce Type: new Abstract: Artificial intelligence (AI) and computer vision are transforming transportation data collection. This study introduces an AI-enabled analytics framework leveraging existing CCTV infrastructure to evaluate the impact of soft...

<https://arxiv.org/abs/2605.05402>

4 CONF

When2Speak: A Dataset for Temporal Participation and Turn-Taking in Multi-Party Conversations for Large Language Models

ArXiv CS.AI 50%

arXiv:2605.05626v1 Announce Type: cross Abstract: Large Language Models (LLMs) excel at generating contextually appropriate responses but remain poorly calibrated for multi-party conversations, where deciding when to speak is as critical as what to say. In such settings, naively...

<https://arxiv.org/abs/2605.05626>

5 CONF

LaTA: A Drop-in, FERPA-Compliant Local-LLM Autograder for Upper-Division STEM Coursework

ArXiv CS.AI 50%

arXiv:2605.05410v1 Announce Type: new Abstract: Large-language-model (LLM) graders promise to relieve the grading burden of upper-division STEM courses, but most deployments to date send student work to third-party APIs, violating FERPA and exposing institutions to data risk...

<https://arxiv.org/abs/2605.05410>

6 CONF

SafeHarbor: Hierarchical Memory-Augmented Guardrail for LLM Agent Safety

ArXiv CS.AI 50%

arXiv:2605.05704v1 Announce Type: cross Abstract: With the rapid evolution of foundation models, Large Language Model (LLM) agents have demonstrated increasingly powerful tool-use capabilities. However, this proficiency introduces significant security risks, as malicious actors...

<https://arxiv.org/abs/2605.05704>

7 CONF

The Geopolitics of AI Safety: A Causal Analysis of Regional LLM Bias

ArXiv CS.AI 50%

arXiv:2605.05427v1 Announce Type: new Abstract: As Large Language Models (LLMs) are integrated into global software systems, ensuring equitable safety guardrails is a critical requirement. Current fairness evaluations predominantly measure bias observationally, a methodology...

<https://arxiv.org/abs/2605.05427>

8 CONF

Authorization Propagation in Multi-Agent AI Systems: Identity Governance as Infrastructure

ArXiv CS.AI 50%

arXiv:2605.05440v1 Announce Type: new Abstract: The security discussion around agentic AI focuses heavily on prompt injection. This paper argues that multi-agent systems also create a distinct authorization problem: maintaining authorization invariants as non-human principals...

<https://arxiv.org/abs/2605.05440>

9 CONF

Instrumental Choices: Measuring the Propensity of LLM Agents to Pursue Instrumental Behaviors

ArXiv CS.AI 50%

arXiv:2605.06490v1 Announce Type: new Abstract: AI systems have become increasingly capable of dangerous behaviours in many domains. This raises the question: Do models sometimes choose to violate human instructions in order to perform behaviour that is more useful for certain...

<https://arxiv.org/abs/2605.06490>

0 CONF

LANTERN: LLM-Augmented Neurosymbolic Transfer with Experience-Gated Reasoning Networks

ArXiv CS.AI 50%

arXiv:2605.05478v1 Announce Type: new Abstract: Transfer learning in reinforcement learning (RL) seeks to accelerate learning in new tasks by leveraging knowledge from related sources. Existing neurosymbolic transfer methods, however, typically rely on manually specified task...

<https://arxiv.org/abs/2605.05478>

1 CONF

FinRAG-12B: A Production-Validated Recipe for Grounded Question Answering in Banking

ArXiv CS.AI 50%

arXiv:2605.05482v1 Announce Type: new Abstract: Large language models (LLMs) are rapidly being adopted across various domains. However, their adoption in banking industry faces resistance due to demands for high accuracy, regulatory compliance, and the need for verifiable and...

<https://arxiv.org/abs/2605.05482>

2 CONF

LLM-AutoDP: Automatic Data Processing via LLM Agents for Model Fine-tuning

ArXiv CS.AI 50%

arXiv:2601.20375v2 Announce Type: replace-cross Abstract: Large Language Models (LLMs) can be fine-tuned on domain-specific data to enhance their performance in specialized fields. However, such data often contains numerous low-quality samples, necessitating effective data...

<https://arxiv.org/abs/2601.20375>

3 CONF

AgenticRAG: Agentic Retrieval for Enterprise Knowledge Bases

ArXiv CS.AI 50%

arXiv:2605.05538v1 Announce Type: new Abstract: We present AgenticRAG, a practical agentic harness for retrieval and analysis over enterprise knowledge bases. Standard RAG pipelines place significant burden of grounding on the search stack, constraining the language model to a...

<https://arxiv.org/abs/2605.05538>

- 4
CONF

AdaGamma: State-Dependent Discounting for Temporal Adaptation in Reinforcement Learning

ArXiv CS.AI
50%

arXiv:2605.06149v1 Announce Type: cross Abstract: The discount factor in reinforcement learning controls both the effective planning horizon and the strength of bootstrapping, yet most deep RL methods use a single fixed value across all states. While state-dependent discounting...

<https://arxiv.org/abs/2605.06149>
- 5
CONF

Causal Probing for Internal Visual Representations in Multimodal Large Language Models

ArXiv CS.AI
50%

arXiv:2605.05593v1 Announce Type: new Abstract: Despite the remarkable success of Multimodal Large Language Models (MLLMs) across diverse tasks, the internal mechanisms governing how they encode and ground distinct visual concepts remain poorly understood. To bridge this gap, we...

<https://arxiv.org/abs/2605.05593>
- 6
CONF

Prober.ai: Gated Inquiry-Based Feedback via LLM-Constrained Personas for Argumentative Writing Development

ArXiv CS.AI
50%

arXiv:2605.05598v1 Announce Type: new Abstract: The proliferation of large language models (LLMs) in educational settings has paradoxically undermined the cognitive processes they purport to support. Students increasingly outsource critical thinking to AI assistants that...

<https://arxiv.org/abs/2605.05598>
- 7
CONF

Text-Graph Synergy: A Bidirectional Verification and Completion Framework for RAG

ArXiv CS.AI
50%

arXiv:2605.05643v1 Announce Type: new Abstract: Retrieval-Augmented Generation (RAG) has become a core paradigm for enhancing factual grounding and multi-hop reasoning in Large Language Models (LLMs). Traditional text-based RAG often retrieves logically irrelevant...

<https://arxiv.org/abs/2605.05643>
- 8
CONF

Are Large Language Models Robust in Understanding Code Against Semantics-Preserving Mutations?

ArXiv CS.AI
50%

arXiv:2505.10443v3 Announce Type: replace-cross Abstract: With the widespread adoption of vibe coding, understanding the reasoning and robustness of Large Language Models (LLMs) is critical for their reliable use in programming tasks. While recent studies assess LLMs' ability to...

<https://arxiv.org/abs/2505.10443>
- 9
CONF

Chain of Risk: Safety Failures in Large Reasoning Models and Mitigation via Adaptive Multi-Principle Steering

ArXiv CS.AI
50%

arXiv:2605.05678v1 Announce Type: new Abstract: Large reasoning models (LRMs) increasingly expose chain-of-thought-like reasoning for transparency, verification, and deliberate problem solving. This creates a safety blind spot: harmful or policy-violating content may appear in...

<https://arxiv.org/abs/2605.05678>

- 0
CONF

Attractor Geometry of Transformer Memory: From Conflict Arbitration to Confident Hallucination

ArXiv CS.AI
50%

arXiv:2605.05686v1 Announce Type: new Abstract: Language models draw on two knowledge sources: facts baked into weights (parametric memory, PM) and information in context (working memory, WM). We study two mechanistically distinct failure modes--conflict, when PM and WM disagree...

<https://arxiv.org/abs/2605.05686>
- 1
CONF

GCCM: Enhancing Generative Graph Prediction via Contrastive Consistency Model

ArXiv CS.AI
50%

arXiv:2605.05689v1 Announce Type: new Abstract: Conditional generative models, particularly diffusion-based methods, have recently been applied to graph prediction by modeling the target as a conditional distribution given the input graph, yielding competitive results compared...

<https://arxiv.org/abs/2605.05689>
- 2
CONF

Inference-Time Budget Control for LLM Search Agents

ArXiv CS.AI
50%

arXiv:2605.05701v1 Announce Type: new Abstract: LLM search agents increasingly rely on tools at inference time, but their trajectories are often constrained by hard limits on both tool calls and generated tokens. Under such dual budgets, better answers require not only stronger...

<https://arxiv.org/abs/2605.05701>
- 3
CONF

Conceal, Reconstruct, Jailbreak: Exploiting the Reconstruction-Concealment Tradeoff in MLLMs

ArXiv CS.AI
50%

arXiv:2605.05709v1 Announce Type: new Abstract: Intent-obfuscation-based jailbreak attacks on multimodal large language models (MLLMs) transform a harmful query into a concealed multimodal input to bypass safety mechanisms. We show that such attacks are governed by a...

<https://arxiv.org/abs/2605.05709>
- 4
CONF

Decodable but Not Corrected by Fixed Residual-Stream Linear Steering: Evidence from Medical LLM Failure Regimes

ArXiv CS.AI
50%

arXiv:2605.05715v1 Announce Type: new Abstract: Can linearly decodable failure signals in LLM hidden states be leveraged to correct those failures? We investigate this classification-correction gap via Overthinking (OT)--a stable behavioral regime (Jaccard ≥ 0.81 , 94%...

<https://arxiv.org/abs/2605.05715>
- 5
CONF

Detecting AI-Generated Videos with Spiking Neural Networks

ArXiv CS.AI
50%

arXiv:2605.05895v1 Announce Type: cross Abstract: Modern AI-generated videos are photorealistic at the single-frame level, leaving inter-frame dynamics as the main remaining axis for detection. Existing detectors typically handle this temporal evidence in three ways: feeding the...

<https://arxiv.org/abs/2605.05895>

6 CONF

SkillRet: A Large-Scale Benchmark for Skill Retrieval in LLM Agents

ArXiv CS.AI 50%

arXiv:2605.05726v1 Announce Type: new Abstract: As LLM agents are increasingly deployed with large libraries of reusable skills, selecting the right skill for a user request has become a critical systems challenge. In small libraries, users may invoke skills explicitly by name,...

<https://arxiv.org/abs/2605.05726>

7 CONF

Knee Osteoarthritis Severity Grading Using Optimized Deep Learning and LLM-Driven Intelligent AI on Computationally Limited Systems

ArXiv CS.AI 50%

arXiv:2605.05731v1 Announce Type: new Abstract: Knee osteoarthritis (KOA) is among the musculoskeletal disorders that considerably restrict joint mobility, cause severe chronic pain and impact negatively on quality life. It is one of the persistent health issues worldwide....

<https://arxiv.org/abs/2605.05731>

8 CONF

ReFlect: An Effective Harness System for Complex Long-Horizon LLM Reasoning

ArXiv CS.AI 50%

arXiv:2605.05737v1 Announce Type: new Abstract: Current reasoning paradigms for LLMs include chain-of-thought, ReAct, and post-hoc self-critique. These paradigms rely on two assumptions that fail on long-horizon, multi-stage tasks. As a result, errors accumulate silently across...

<https://arxiv.org/abs/2605.05737>

9 CONF

StraTA: Incentivizing Agentic Reinforcement Learning with Strategic Trajectory Abstraction

ArXiv CS.AI 50%

arXiv:2605.06642v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly used as interactive agents, but optimizing them for long-horizon decision making remains difficult because current methods are largely purely reactive, which weakens both exploration...

<https://arxiv.org/abs/2605.06642>

0 CONF

Evaluating Explainability in Safety-Critical ATR Systems: Limitations of Post-Hoc Methods and Paths Toward Robust XAI

ArXiv CS.AI 50%

arXiv:2605.05748v1 Announce Type: new Abstract: Explainable Artificial Intelligence (XAI) is increasingly recognized as essential for deploying machine learning systems in safety critical environments. In Automatic Target Recognition (ATR), where models operate on image, video,...

<https://arxiv.org/abs/2605.05748>

1 CONF

Confidence is the key: how conformal prediction enhances the generative design of permeable peptides

ArXiv CS.AI 50%

arXiv:2605.05770v1 Announce Type: new Abstract: Generative models coupled with reinforcement learning (RL), such as REINVENT and PepINVENT, have emerged as a powerful framework for de novo molecular design. During the ideation process these generative frameworks utilize various...

<https://arxiv.org/abs/2605.05770>

- 2 **CONF**
CircuitFormer: A Circuit Language Model for Analog Topology Design from Natural Language Prompt
ArXiv CS.AI 50%
arXiv:2605.05773v1 Announce Type: new Abstract: Automating analog circuit design remains a longstanding challenge in Electronic Design Automation (EDA). While Transformer-based Large Language Models (LLMs) have revolutionized software code generation, their application to analog...
<https://arxiv.org/abs/2605.05773>
- 3 **CONF**
HEDP: A Hybrid Energy-Distance Prompt-based Framework for Domain Incremental Learning
ArXiv CS.AI 50%
arXiv:2605.05776v1 Announce Type: new Abstract: Domain Incremental Learning is a critical scenario that requires models to continuously adapt to new data domains without retraining. However, domain shifts often cause severe performance degradation. To address this, we propose...
<https://arxiv.org/abs/2605.05776>
- 4 **CONF**
DeEscalWild: A Real-World Benchmark for Automated De-Escalation Training with SLMs
ArXiv CS.AI 50%
arXiv:2604.13075v2 Announce Type: replace-cross Abstract: Effective de-escalation is critical for law enforcement safety and community trust, yet traditional training methods lack scalability and realism. While Large Language Models (LLMs) enable dynamic, open-ended simulations,...
<https://arxiv.org/abs/2604.13075>
- 5 **CONF**
Taklif.AI: LLM-Powered Platform for Interest-Based Personalized College Assignments
ArXiv CS.AI 50%
arXiv:2605.05842v1 Announce Type: new Abstract: Educators face significant challenges in creating engaging, personalized assignments that accommodate students' diverse interests and cognitive abilities. Traditional one-size-fits-all assignments frequently lead to decreased...
<https://arxiv.org/abs/2605.05842>
- 6 **CONF**
Event-Causal RAG: A Retrieval-Augmented Generation Framework for Long Video Reasoning in Complex Scenarios
ArXiv CS.AI 50%
arXiv:2605.06185v1 Announce Type: new Abstract: Recent large vision-language models have achieved strong performance on short- and medium-length video understanding, yet they remain inadequate for ultra-long or even infinite video reasoning, where models must preserve coherent...
<https://arxiv.org/abs/2605.06185>
- 7 **CONF**
Dynamic Expert-Guided Model Averaging for Causal Discovery
ArXiv CS.AI 50%
arXiv:2601.16715v2 Announce Type: replace-cross Abstract: Would-be practitioners of causal discovery face a dizzying array of algorithms without a clear best choice. This abundance of competitive methods makes ensembling a natural strategy for practical applications. At the same...
<https://arxiv.org/abs/2601.16715>

- 8 **CONF**
- From Coordinate Matching to Structural Alignment: Rethinking Prototype Alignment in Heterogeneous Federated Learning**
- ArXiv CS.AI 50%
- arXiv:2605.05959v1 Announce Type: new Abstract: Heterogeneous federated learning (HtFL) aims to enable collaboration among clients that differ in both data distributions and model architectures. Prototype-based methods, which communicate class-level feature centers (prototypes)...
- <https://arxiv.org/abs/2605.05959>
-
- 9 **CONF**
- Strat-LLM: Stratified Strategy Alignment for LLM-based Stock Trading with Real-time Multi-Source Signals**
- ArXiv CS.AI 50%
- arXiv:2605.06024v1 Announce Type: new Abstract: Large Language Models (LLMs) are evolving into autonomous trading agents, yet existing benchmarks often overlook the interplay between architectural reasoning and strategy consistency. We propose Strat-LLM, a framework grounded in...
- <https://arxiv.org/abs/2605.06024>
-
- 0 **CONF**
- Novelty-based Tree-of-Thought Search for LLM Reasoning and Planning**
- ArXiv CS.AI 50%
- arXiv:2605.06040v1 Announce Type: new Abstract: Although advances such as chain-of-thought, tree-of-thought or reinforcement learning have improved the performance of LLMs in reasoning and planning tasks, they are still brittle and have not achieved human-level performance in...
- <https://arxiv.org/abs/2605.06040>
-
- 1 **CONF**
- Visual Fingerprints for LLM Generation Comparison**
- ArXiv CS.AI 50%
- arXiv:2605.06054v1 Announce Type: new Abstract: Large language model (LLM) outputs arise from complex interactions among prompts, system instructions, model parameters, and architecture. We refer to specific configurations of these factors as generation conditions, each of which...
- <https://arxiv.org/abs/2605.06054>
-
- 2 **CONF**
- VibeServe: Can AI Agents Build Bespoke LLM Serving Systems?**
- ArXiv CS.AI 50%
- arXiv:2605.06068v1 Announce Type: new Abstract: For years, we have built LLM serving systems like any other critical infrastructure: a single general-purpose stack, hand-tuned over many engineer-years, meant to support every model and workload. In this paper, we take the...
- <https://arxiv.org/abs/2605.06068>
-
- 3 **CONF**
- Shallow Prefill, Deep Decoding: Efficient Long-Context Inference via Layer-Asymmetric KV Visibility**
- ArXiv CS.AI 50%
- arXiv:2605.06105v1 Announce Type: new Abstract: Long-context inference in decoder-only language models is costly because long prompts are processed during Prefill, cached at every layer, and repeatedly attended to during autoregressive Decode. We introduce \emph{Shallow Prefill},...
- <https://arxiv.org/abs/2605.06105>

- 4
CONF

Rethinking Data Curation in LLM Training: Online Reweighting Offers Better Generalization than Offline Methods

ArXiv CS.AI 50%

arXiv:2605.05227v1 Announce Type: cross Abstract: Data curation is a critical yet under-explored area in large language model (LLM) training. Existing methods, such as data selection and mixing, operate in an offline paradigm, detaching themselves from training. This separation...

<https://arxiv.org/abs/2605.05227>
- 5
CONF

Back to the Beginning of Heuristic Design: Bridging Code and Knowledge with LLMs

ArXiv CS.AI 50%

arXiv:2605.06123v1 Announce Type: new Abstract: Large language models (LLMs) have recently advanced automatic heuristic design (AHD) for combinatorial optimization (CO), where candidate heuristics are iteratively proposed, evaluated, and refined. Most existing approaches search...

<https://arxiv.org/abs/2605.06123>
- 6
CONF

P-Guide: Parameter-Efficient Prior Steering for Single-Pass CFG Inference

ArXiv CS.AI 50%

arXiv:2605.06124v1 Announce Type: new Abstract: Classifier-Free Guidance (CFG) is essential for high-fidelity conditional generation in flow matching, yet it imposes significant computational overhead by requiring dual forward passes at each sampling step. In this work, we...

<https://arxiv.org/abs/2605.06124>
- 7
CONF

TFM-Retouche: A Lightweight Input-Space Adapter for Tabular Foundation Models

ArXiv CS.AI 50%

arXiv:2605.06047v1 Announce Type: cross Abstract: Tabular foundation models (TFMs), such as TabPFN-2.6, TabICLv2, ConTextTab, Mitra, LimiX, and TabDPT, achieve strong zero-shot performance through in-context learning, but their inductive biases remain fixed at inference time....

<https://arxiv.org/abs/2605.06047>
- 8
CONF

Beyond Accuracy: Policy Invariance as a Reliability Test for LLM Safety Judges

ArXiv CS.AI 50%

arXiv:2605.06161v1 Announce Type: new Abstract: LLM-as-a-Judge pipelines have become the de facto evaluator for agent safety, yet existing benchmarks treat their verdicts as ground-truth proxies without checking whether the verdicts depend on the agent's behavior or merely on...

<https://arxiv.org/abs/2605.06161>
- 9
CONF

Beyond Fixed Benchmarks and Worst-Case Attacks: Dynamic Boundary Evaluation for Language Models

ArXiv CS.AI 50%

arXiv:2605.06213v1 Announce Type: new Abstract: Evaluating large language models (LLMs) today rests on fixed benchmarks that apply the same set of items to any model, producing ceiling and floor effects that mask capability gaps. We argue that the most informative evaluation...

<https://arxiv.org/abs/2605.06213>

0 **CONF****SOWing Information: Cultivating Contextual Coherence with MLLMs in Image Generation****ArXiv CS.AI 50%**

arXiv:2411.19182v2 Announce Type: replace-cross Abstract: Originating from the diffusion phenomenon in physics, which describes the random movement and collisions of particles, diffusion generative models simulate a random walk in the data space along the denoising trajectory....

<https://arxiv.org/abs/2411.19182>

1 **CONF****Retina-RAG: Retrieval-Augmented Vision-Language Modeling for Joint Retinal Diagnosis and Clinical Report Generation****ArXiv CS.AI 50%**

arXiv:2605.06173v1 Announce Type: cross Abstract: Diabetic Retinopathy (DR) is a leading cause of preventable blindness among working-age adults worldwide, yet most automated screening systems are limited to image-level classification and lack clinically structured reporting. We...

<https://arxiv.org/abs/2605.06173>

2 **CONF****Price of Fairness in Short-Term and Long-Term Algorithmic Selections****ArXiv CS.AI 50%**

arXiv:2605.06227v1 Announce Type: new Abstract: Algorithmic decision-making in high-stakes settings can have profound impacts on individuals and populations. While much prior work studies fairness in static settings, recent results show that enforcing static fairness constraints...

<https://arxiv.org/abs/2605.06227>

3 **CONF****Addressing Labelled Data Scarcity: Taxonomy-Agnostic Annotation of PII Values in HTTP Traffic using LLMs****ArXiv CS.AI 50%**

arXiv:2605.06305v1 Announce Type: new Abstract: Automated privacy audits of web and mobile applications often analyse outbound HTTP traffic to detect Personally Identifiable Information (PII) leakage. However, existing learning-based detectors typically depend on scarce...

<https://arxiv.org/abs/2605.06305>

4 **CONF****Measuring Black-Box Confidence via Reasoning Trajectories: Geometry, Coverage, and Verbalization****ArXiv CS.AI 50%**

arXiv:2605.06308v1 Announce Type: new Abstract: Reliable confidence estimation enables safe deployment of chain-of-thought (CoT) reasoning through text-only APIs. Yet the dominant black-box baseline, self-consistency over K samples, is linearly expensive and ignores the geometry...

<https://arxiv.org/abs/2605.06308>

5 **CONF****Mind the Gap? A Distributional Comparison of Real and Synthetic Priors for Tabular Foundation Models****ArXiv CS.AI 50%**

arXiv:2605.06343v1 Announce Type: new Abstract: Tabular foundation models are pre-trained on one of three classes of corpus: curated datasets drawn from benchmark repositories, tables harvested at scale from the web, or synthetic tables sampled from a parametric generative...

<https://arxiv.org/abs/2605.06343>

6 CONF

SynBench: A Benchmark for Differentially Private Text Generation

ArXiv CS.AI 50%

arXiv:2509.14594v2 Announce Type: replace Abstract: Synthetic text generation with Differential Privacy (DP) guarantees emerges as a principled approach that can enable the sharing of sensitive datasets across institutional and regulatory boundaries, while bounding the risks of...

<https://arxiv.org/abs/2509.14594>

7 CONF

From Agent Loops to Deterministic Graphs: Execution Lineage for Reproducible AI-Native Work

ArXiv CS.AI 50%

arXiv:2605.06365v1 Announce Type: new Abstract: Large language model systems are increasingly deployed as agentic workflows that interleave reasoning, tool use, memory, and iterative refinement. These systems are effective at producing answers, but they often rely on implicit...

<https://arxiv.org/abs/2605.06365>

8 CONF

Debiased Multimodal Personality Understanding through Dual Causal Intervention

ArXiv CS.AI 50%

arXiv:2605.06371v1 Announce Type: new Abstract: Multimodal personality understanding plays a critical role in human centered artificial intelligence. Previous work mainly focus on learning rich multimodal representations for video personality understanding. However, they often suffer...

<https://arxiv.org/abs/2605.06371>

9 CONF

Rethinking Vacuity for OOD Detection in Evidential Deep Learning

ArXiv CS.AI 50%

arXiv:2605.06382v1 Announce Type: new Abstract: Vacuity, or Uncertainty Mass (UM), is commonly used as a metric to evaluate Out-of-Distribution (OOD) detection in Evidential Deep Learning (EDL). It generally involves dividing the number of classes (K) by the total strength of...

<https://arxiv.org/abs/2605.06382>

10 CONF

Automated alignment is harder than you think

ArXiv CS.AI 50%

arXiv:2605.06390v1 Announce Type: new Abstract: A leading proposal for aligning artificial superintelligence (ASI) is to use AI agents to automate an increasing fraction of alignment research as capabilities improve. We argue that, even when research agents are not scheming to...

<https://arxiv.org/abs/2605.06390>

1 CONF

Knowledge Graphs, the Missing Link in Agentic AI-based Formal Verification

ArXiv CS.AI 50%

arXiv:2605.06434v1 Announce Type: new Abstract: Recent advances in Large Language Models (LLMs) have enabled workflows that generate SystemVerilog Assertions (SVAs) from natural-language specifications, with the potential to accelerate Formal Verification (FV). However,...

<https://arxiv.org/abs/2605.06434>

2 CONF

PrefixGuard: From LLM-Agent Traces to Online Failure-Warning Monitors

ArXiv CS.AI 50%

arXiv:2605.06455v1 Announce Type: new Abstract: Large language model (LLM) agents now execute long, tool-using tasks where final outcome checks can arrive too late for intervention. Online warning requires lightweight prefix monitors over heterogeneous traces, but hand-authored...

<https://arxiv.org/abs/2605.06455>

3 **CONF****Patch-Effect Graph Kernels for LLM Interpretability**

ArXiv CS.AI 50%

arXiv:2605.06480v1 Announce Type: new Abstract: Mechanistic interpretability aims to reverse-engineer transformer computations by identifying causal circuits through activation patching. However, scaling these interventions across diverse prompts and task families produces...

<https://arxiv.org/abs/2605.06480>
4 **CONF****Ex Ante Evaluation of AI-Induced Idea Diversity Collapse**

ArXiv CS.AI 50%

arXiv:2605.06540v1 Announce Type: new Abstract: Creative AI systems are typically evaluated at the level of individual utility, yet creative outputs are consumed in populations: an idea loses value when many others produce similar ones. This creates an evaluation blind spot, as...

<https://arxiv.org/abs/2605.06540>
5 **CONF****Improved techniques for fine-tuning flow models via adjoint matching: a deterministic control pipeline**

ArXiv CS.AI 50%

arXiv:2605.06583v1 Announce Type: new Abstract: We propose a deterministic adjoint matching framework that formulates human preference alignment for flow-based generative models as an optimal control problem over velocity fields. One can directly regress the control toward a...

<https://arxiv.org/abs/2605.06583>
6 **CONF****Active Learning for Communication Structure Optimization in LLM-Based Multi-Agent Systems**

ArXiv CS.AI 50%

arXiv:2605.05703v1 Announce Type: cross Abstract: Optimizing the communication structure of large language model based multi-agent systems (LLM-MAS) has been shown to improve downstream performance and reduce token usage. Existing methods typically rely on randomly sampled...

<https://arxiv.org/abs/2605.05703>
7 **CONF****Information Aggregation with AI Agents**

ArXiv CS.AI 50%

arXiv:2604.20050v2 Announce Type: replace-cross Abstract: Can Large Language Models (AI agents) aggregate dispersed private information through trading and reason about the knowledge of others by observing price movements? We conduct a controlled experiment where AI agents trade...

<https://arxiv.org/abs/2604.20050>
8 **CONF****Physics-Informed Neural Networks with Learnable Loss Balancing and Transfer Learning**

ArXiv CS.AI 50%

arXiv:2605.05217v1 Announce Type: cross Abstract: We propose a self-supervised physics-informed neural network (PINN) framework that adaptively balances physics-based and data-driven supervision for scientific machine learning under data scarcity. Unlike prior PINNs that rely on...

<https://arxiv.org/abs/2605.05217>

9 CONF

Auction-Based Regulation for Artificial Intelligence

ArXiv CS.AI 50%

arXiv:2410.01871v3 Announce Type: replace-cross Abstract: In an era of "moving fast and breaking things", regulators have moved slowly to pick up the safety, bias, and legal debris left in the wake of broken Artificial Intelligence (AI) deployment. While there is much-warranted...

<https://arxiv.org/abs/2410.01871>

0 CONF

Sparse Prefix Caching for Hybrid and Recurrent LLM Serving

ArXiv CS.AI 50%

arXiv:2605.05219v1 Announce Type: cross Abstract: Prefix caching is a key latency optimization for autoregressive LLM serving, yet existing systems assume dense per-token key/value reuse. State-space models change the structure of the problem: a recurrent layer can resume from a...

<https://arxiv.org/abs/2605.05219>

1 CONF

Beyond Autoregressive RTG: Conditioning via Injection Outside Sequential Modeling in Decision Transformer

ArXiv CS.AI 50%

arXiv:2605.06104v1 Announce Type: cross Abstract: Decision Transformer (DT) formulates offline reinforcement learning as autoregressive sequence modeling, achieving promising results by predicting actions from a sequence of Return-to-Go (RTG), state, and action tokens. However,...

<https://arxiv.org/abs/2605.06104>

2 CONF

Channel-Level Semantic Perturbations: Unlearnable Examples for Diverse Training Paradigms

ArXiv CS.AI 50%

arXiv:2605.05224v1 Announce Type: cross Abstract: The unauthorized use of personal data in model training has emerged as a growing privacy threat. Unlearnable examples (UEs) address this issue by embedding imperceptible perturbations into benign examples to obstruct feature...

<https://arxiv.org/abs/2605.05224>

3 CONF

MOSAIC: Module Discovery via Sparse Additive Identifiable Causal Learning for Scientific Time Series

ArXiv CS.AI 50%

arXiv:2605.05524v1 Announce Type: cross Abstract: Causal representation learning (CRL) seeks to recover latent variables with identifiability guarantees, typically up to permutation and component-wise reparameterization under appropriate assumptions. However, identifiability...

<https://arxiv.org/abs/2605.05524>

4 CONF

asRoBallet: Closing the Sim2Real Gap via Friction-Aware Reinforcement Learning for Underactuated Spherical Dynamics

ArXiv CS.AI 50%

arXiv:2604.24916v2 Announce Type: replace-cross Abstract: We introduce asRoBallet, to the best of our knowledge, the first end-to-end reinforcement learning (RL) locomotion policy deployed on a humanoid ballbot hardware platform. Historically, ballbots have served as a canonical...

<https://arxiv.org/abs/2604.24916>

5 CONF

Schedule-and-Calibrate: Utility-Guided Multi-Task Reinforcement Learning for Code LLMs

ArXiv CS.AI 50%

arXiv:2605.06111v1 Announce Type: cross Abstract: Reinforcement learning (RL) with verifiable rewards has proven effective at post-training LLMs for coding, yet deploying separate task-specific specialists incurs costs that scale with the number of tasks, motivating a unified...

<https://arxiv.org/abs/2605.06111>

6 CONF

Automated Population-Level Audit Assurance via AI-Based Document Intelligence

ArXiv CS.AI 50%

arXiv:2605.05252v1 Announce Type: cross Abstract: Audit transaction testing validates accuracy and completeness of customer-facing statements against internal systems of record. Traditional manual, sample-based review of unstructured PDF statements is labor-intensive and does...

<https://arxiv.org/abs/2605.05252>

7 CONF

Career-Aware Resume Tailoring via Multi-Source Retrieval-Augmented Generation with Provenance Tracking: A Case Study

ArXiv CS.AI 50%

arXiv:2605.05257v1 Announce Type: cross Abstract: AI-assisted resume tailoring systems commonly operate on a single uploaded resume, which limits their ability to recover relevant experience omitted from the current draft and makes it difficult for users to distinguish grounded...

<https://arxiv.org/abs/2605.05257>

8 CONF

Bridging Generation and Training: A Systematic Review of Quality Issues in LLMs for Code

ArXiv CS.AI 50%

arXiv:2605.05267v1 Announce Type: cross Abstract: Large language models (LLMs) frequently generate defective outputs in code generation tasks, ranging from logical bugs to security vulnerabilities. While these generation failures are often treated as model-level limitations,...

<https://arxiv.org/abs/2605.05267>

9 CONF

Shattering the Echo Chamber: Hidden Safeguards in Manuscripts Against the AI Takeover of Peer Review

ArXiv CS.AI 50%

arXiv:2605.05271v1 Announce Type: cross Abstract: As LLMs become increasingly capable, editorial boards and program committees are growing concerned about reviewers who fully outsource peer review to commercial chatbots. This concern stems from prior findings that current...

<https://arxiv.org/abs/2605.05271>

00 CONF

Open-SAT: LLM-Guided Query Embedding Refinement for Open-Vocabulary Object Retrieval in Satellite Imagery

ArXiv CS.AI 50%

arXiv:2605.05344v1 Announce Type: cross Abstract: In satellite applications, user queries often take the form of open-ended natural language, extending beyond a fixed set of predefined categories. This open-vocabulary nature poses significant challenges for retrieving relevant...

<https://arxiv.org/abs/2605.05344>

01 CONF

Making AI Drafts Count: A Quality Threshold in Audio Description Workflows

ArXiv CS.AI 50%

arXiv:2605.05348v1 Announce Type: cross Abstract: Audio description (AD) narrates visual elements in video for blind and low-vision audiences. Recent work has shown that giving novice describers an AI-generated draft to start from helps produce higher-quality AD and lowers the...

<https://arxiv.org/abs/2605.05348>

02 CONF

Counterargument for Critical Thinking as Judged by AI and Humans

ArXiv CS.AI 50%

arXiv:2605.05353v1 Announce Type: cross Abstract: This intervention study investigates the use of counterarguments in writing for critical thinking by students in the context of Generative AI (GenAI). This is especially as risks of cheating and cognitive offloading exist with...

<https://arxiv.org/abs/2605.05353>

03 CONF

LiteSpark Inference on Consumer CPUs: Custom SIMD Kernels for Ternary Neural Networks

ArXiv CS.AI 50%

arXiv:2605.06485v1 Announce Type: cross Abstract: Large language models (LLMs) have transformed artificial intelligence, but their computational requirements remain prohibitive for most users. Standard inference demands expensive datacenter GPUs or cloud API access, leaving over...

<https://arxiv.org/abs/2605.06485>

04 CONF

On Semantic Loss Fine-Tuning Approach for Preventing Model Collapse in Causal Reasoning

ArXiv CS.AI 50%

arXiv:2605.05438v1 Announce Type: cross Abstract: Standard fine-tuning of transformer models on causal reasoning tasks leads to catastrophic model collapse, where models learn trivial solutions such as always predicting "Yes" or "No" regardless of input structure. We demonstrate...

<https://arxiv.org/abs/2605.05438>

05 CONF

SLAM: Structural Linguistic Activation Marking for Language Models

ArXiv CS.AI 50%

arXiv:2605.05443v1 Announce Type: cross Abstract: LLM watermarks must be detectable without compromising text quality, yet most existing schemes bias the next-token distribution and pay for detection with measurable quality loss. We present SLAM (Structural Linguistic Activation...

<https://arxiv.org/abs/2605.05443>

06 CONF

The Pedagogy of AI Mistakes: Fostering Higher-Order Thinking

ArXiv CS.AI 50%

arXiv:2605.05472v1 Announce Type: cross Abstract: As generative AI becomes increasingly integrated into higher education, its frequent errors and hallucinations, often seen as limitations, offer a unique pedagogical opportunity. By framing AI as a "learning companion" whose...

<https://arxiv.org/abs/2605.05472>

07 CONF

ReaComp: Compiling LLM Reasoning into Symbolic Solvers for Efficient Program Synthesis

ArXiv CS.AI 50%

arXiv:2605.05485v1 Announce Type: cross Abstract: LLMs can solve program synthesis tasks but remain inefficient and unreliable on hard instances requiring large combinatorial search. Given a small set of reasoning traces, we use coding agents to compile them into reusable...

<https://arxiv.org/abs/2605.05485>

08 CONF

AstroAlertBench: Evaluating the Accuracy, Reasoning, and Honesty of Multimodal LLMs in Astronomical Classification

ArXiv CS.AI 50%

arXiv:2605.05573v1 Announce Type: cross Abstract: Modern astronomical observatories generate a massive volume of multimodal data, creating a critical bottleneck for expert human review. While multimodal large language models (LLMs) have shown promise in interpreting complex...

<https://arxiv.org/abs/2605.05573>

09 CONF

Time Series Reasoning via Process-Verifiable Thinking Data Synthesis and Scheduling for Tailored LLM Reasoning

ArXiv CS.AI 50%

arXiv:2602.07830v2 Announce Type: replace Abstract: Time series is a pervasive data type across various application domains, rendering the reasonable solving of diverse time series tasks a long-standing goal. Recent advances in large language models (LLMs), especially their...

<https://arxiv.org/abs/2602.07830>

10 CONF

One Turn Too Late: Response-Aware Defense Against Hidden Malicious Intent in Multi-Turn Dialogue

ArXiv CS.AI 50%

arXiv:2605.05630v1 Announce Type: cross Abstract: Hidden malicious intent in multi-turn dialogue poses a growing threat to deployed large language models (LLMs). Rather than exposing a harmful objective in a single prompt, increasingly capable attackers can distribute their...

<https://arxiv.org/abs/2605.05630>

11 CONF

The Missing Evaluation Axis: What 10,000 Student Submissions Reveal About AI Tutor Effectiveness

ArXiv CS.AI 50%

arXiv:2605.05648v1 Announce Type: cross Abstract: Current Artificial Intelligence (AI)-based tutoring systems (AI tutors) are primarily evaluated based on the pedagogical quality of their feedback messages. While important, pedagogy alone is insufficient because it ignores a...

<https://arxiv.org/abs/2605.05648>

12 CONF

EGA: Adapting Frozen Encoders for Vector Search with Bounded Out-of-Distribution Degradation

ArXiv CS.AI 50%

arXiv:2605.05674v1 Announce Type: cross Abstract: Vector search systems built on frozen vision encoders face queries from unseen classes at deployment, yet existing adapter training collapses under this shift: high-capacity adapters with global contrastive losses silently...

<https://arxiv.org/abs/2605.05674>

13 CONF

Temporal Functional Circuits: From Spline Plots to Faithful Explanations in KAN Forecasting

ArXiv CS.AI 50%

arXiv:2605.05685v1 Announce Type: cross Abstract: Unlike MLPs, Kolmogorov-Arnold Networks (KANs) expose explicit learnable edge functions on every connection, enabling mechanistic explanation in time-series forecasting. This paper introduces Temporal Functional Circuits, a...

<https://arxiv.org/abs/2605.05685>

14 CONF

CFE-PPAR: Compression-friendly encryption for privacy-preserving action recognition leveraging video transformers

ArXiv CS.AI 50%

arXiv:2605.05692v1 Announce Type: cross Abstract: Privacy-preserving action recognition (PPAR) enables machines to understand human activities in videos without revealing sensitive visual content. Among the various strategies for PPAR, encryption-based methods achieve strong...

<https://arxiv.org/abs/2605.05692>

15 CONF

Budgeted Attention Allocation: Cost-Conditioned Compute Control for Efficient Transformers

ArXiv CS.AI 50%

arXiv:2605.05697v1 Announce Type: cross Abstract: Transformers usually expose one inference cost per trained model, while deployed systems often need multiple cost-quality operating points. We study Budgeted Attention Allocation, a monotone head-gating mechanism conditioned on a...

<https://arxiv.org/abs/2605.05697>

16 CONF

Auto Research with Specialist Agents Develops Effective and Non-Trivial Training Recipes

ArXiv CS.AI 50%

arXiv:2605.05724v1 Announce Type: cross Abstract: We study auto research as a closed empirical loop driven by external measurement. Each submitted trial carries a hypothesis, an executable code edit, an evaluator-owned outcome, and feedback that shapes the next proposal. The...

<https://arxiv.org/abs/2605.05724>

17 CONF

WARP: A Benchmark for Primal-Dual Warm-Starting of Interior-Point Solvers

ArXiv CS.AI 50%

arXiv:2605.05728v1 Announce Type: cross Abstract: Solving AC Optimal Power Flow (AC-OPF) is of central importance in electricity market operations, where interior-point methods (IPMs) such as IPOPT are the standard solvers. A growing body of work uses machine learning to predict...

<https://arxiv.org/abs/2605.05728>

18 CONF

Separation Assurance between Heterogeneous Fleets of Small Unmanned Aerial Systems via Multi-Agent Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2605.01041v2 Announce Type: replace-cross Abstract: In the envisioned future dense urban airspace, multiple companies will operate heterogeneous fleets of small unmanned aerial systems (sUASs), where each fleet includes several homogeneous aircraft with identical policies...

<https://arxiv.org/abs/2605.01041>

19 CONF

Transformers Provably Implement In-Context Reinforcement Learning with Policy Improvement

ArXiv CS.AI 50%

arXiv:2605.05755v1 Announce Type: cross Abstract: We investigate the ability of transformers to perform in-context reinforcement learning (ICRL), where a model must infer and execute learning algorithms from trajectory data without parameter updates. We show that a linear...

<https://arxiv.org/abs/2605.05755>

20 CONF

Revealing Modular Gradient Noise Imbalance in LLMs: Calibrating Adam via Signal-to-Noise Ratio

ArXiv CS.AI 50%

arXiv:2605.05794v1 Announce Type: cross Abstract: The impressive performance of large language models (LLMs) arises from their massive scale and heterogeneous module composition. However, this structural heterogeneity introduces additional optimization challenges. While adaptive...

<https://arxiv.org/abs/2605.05794>

21 CONF

LeakDojo: Decoding the Leakage Threats of RAG Systems

ArXiv CS.AI 50%

arXiv:2605.05818v1 Announce Type: cross Abstract: Retrieval-Augmented Generation (RAG) enables large language models (LLMs) to leverage external knowledge, but also exposes valuable RAG databases to leakage attacks. As RAG systems grow more complex and LLMs exhibit stronger...

<https://arxiv.org/abs/2605.05818>

22 CONF

Claw-Eval: Towards Trustworthy Evaluation of Autonomous Agents

ArXiv CS.AI 50%

arXiv:2604.06132v3 Announce Type: replace Abstract: Large language models are increasingly deployed as autonomous agents for multi-step workflows in real-world software environments. However, existing agent benchmarks are limited by trajectory-opaque grading, underspecified...

<https://arxiv.org/abs/2604.06132>

23 CONF

CITE: Anytime-Valid Statistical Inference in LLM Self-Consistency

ArXiv CS.AI 50%

arXiv:2605.05873v1 Announce Type: cross Abstract: Large language models often improve reasoning by sampling multiple outputs and aggregating their final answers, but precise and efficient control of error levels remains a challenging task. In particular, deciding when to stop...

<https://arxiv.org/abs/2605.05873>

24 CONF

DBMSolver: A Training-free Diffusion Bridge Sampler for High-Quality Image-to-Image Translation

ArXiv CS.AI 50%

arXiv:2605.05889v1 Announce Type: cross Abstract: Diffusion-based image-to-image (I2I) translation excels in high-fidelity generation but suffers from slow sampling in state-of-the-art Diffusion Bridge Models (DBMs), often requiring dozens of function evaluations (NFEs). We...

<https://arxiv.org/abs/2605.05889>

25 CONF

Logic-Regularized Verifier Elicits Reasoning from LLMs

ArXiv CS.AI 50%

arXiv:2605.05893v1 Announce Type: cross Abstract: Verifiers are crucial components for enhancing modern LLMs' reasoning capability. Typical verifiers require resource-intensive supervised dataset construction, which is costly and faces limitations in data diversity. In this paper,...

<https://arxiv.org/abs/2605.05893>

26 CONF

Architecture-agnostic Lipschitz-constant Bayesian header and its application to resolve semantically proximal classification errors with vision transformers

ArXiv CS.AI 50%

arXiv:2605.05908v1 Announce Type: cross Abstract: Label noise remains a critical bottleneck for the generalization of supervised deep learning models, particularly when errors are structured rather than random. Standard robust training methods often fail in the presence of such...

<https://arxiv.org/abs/2605.05908>

27 CONF

Quantum-enhanced Large Language Models on Quantum Hardware via Cayley Unitary Adapters

ArXiv CS.AI 50%

arXiv:2605.05914v1 Announce Type: cross Abstract: Large language models (LLMs) have transformed artificial intelligence, yet classical architectures impose a fundamental constraint: every trainable parameter demands classical memory that scales unfavourably with model size....

<https://arxiv.org/abs/2605.05914>

28 CONF

LLM-Driven Design Space Exploration of FPGA-based Accelerators

ArXiv CS.AI 50%

arXiv:2605.05920v1 Announce Type: cross Abstract: Designing field-programmable gate array (FPGA)-based accelerators for modern artificial intelligence workloads requires navigating a large and complex hardware design space encompassing architectural parameters, dataflow...

<https://arxiv.org/abs/2605.05920>

29 CONF

Towards Reliable LLM Evaluation: Correcting the Winner's Curse in Adaptive Benchmarking

ArXiv CS.AI 50%

arXiv:2605.05973v1 Announce Type: cross Abstract: Adaptive prompt and program search makes LLM evaluation selection-sensitive. Once benchmark items are reused inside tuning, the observed winner's score need not estimate the fresh-data performance of the full tune-then-deploy...

<https://arxiv.org/abs/2605.05973>

30 CONF

PragLocker: Protecting Agent Intellectual Property in Untrusted Deployments via Non-Portable Prompts

ArXiv CS.AI 50%

arXiv:2605.05974v1 Announce Type: cross Abstract: LLM agents rely on prompts to implement task-specific capabilities based on foundation LLMs, making agent prompts valuable intellectual property. However, in untrusted deployments, adversaries can copy and reuse these prompts...

<https://arxiv.org/abs/2605.05974>

31 CONF

Safety Anchor: Defending Harmful Fine-tuning via Geometric Bottlenecks

ArXiv CS.AI 50%

arXiv:2605.05995v1 Announce Type: cross Abstract: The safety alignment of Large Language Models (LLMs) remains vulnerable to Harmful Fine-tuning (HFT). While existing defenses impose constraints on parameters, gradients, or internal representations, we observe that they can be...

<https://arxiv.org/abs/2605.05995>

32 CONF

A Fine-Grained Understanding of Uniform Convergence for Halfspaces

ArXiv CS.AI 50%

arXiv:2605.06004v1 Announce Type: cross Abstract: We study the fine-grained uniform convergence behavior of halfspaces beyond worst-case VC bounds. For inhomogeneous halfspaces in $\mathbb{R}^d$ with $d \geq 2$, we show that standard first-order VC bounds are essentially tight:...

<https://arxiv.org/abs/2605.06004>

33 CONF

T2I-VerW: Part-level Fine-grained Perception for Text-to-Image Vehicle Retrieval

ArXiv CS.AI 50%

arXiv:2605.06012v1 Announce Type: cross Abstract: Vehicle Re-identification (Re-ID) aims to retrieve the most similar image to a given query from images captured by non-overlapping cameras. Extending vehicle Re-ID from image-only queries to text-based queries enables retrieval...

<https://arxiv.org/abs/2605.06012>

34 CONF

Does Synthetic Data Help? Empirical Evidence from Deep Learning Time Series Forecasters

ArXiv CS.AI 50%

arXiv:2605.06032v1 Announce Type: cross Abstract: Synthetic data has transformed language model training, yet its role in time series forecasting remains poorly understood. We present a large-scale empirical study: nine experiment groups, 4,218 runs systematically evaluating...

<https://arxiv.org/abs/2605.06032>

35 CONF

When AI Meets Science: Research Diversity, Interdisciplinarity, Visibility, and Retractions across Disciplines in a Global Surge

ArXiv CS.AI 50%

arXiv:2605.06033v1 Announce Type: cross Abstract: The extent to which Artificial Intelligence (AI) can trigger generalized paradigm shifts in science is unclear. Although some of these technologies have revolutionized data collection and analysis in specific scientific fields...

<https://arxiv.org/abs/2605.06033>

36 CONF

Optimal Transport for LLM Reward Modeling from Noisy Preference

ArXiv CS.AI 50%

arXiv:2605.06036v1 Announce Type: cross Abstract: Reward models are fundamental to Reinforcement Learning from Human Feedback (RLHF), yet real-world datasets are inevitably corrupted by noisy preference. Conventional training objectives tend to overfit these errors, while...

<https://arxiv.org/abs/2605.06036>

37 CONF

Alternating Reinforcement Learning with Contextual Rubric Rewards: Beyond the Scalarization Strategy

ArXiv CS.AI 50%

arXiv:2603.15646v2 Announce Type: replace-cross Abstract: Reinforcement Learning with Rubric Rewards (RLRR) is a framework that extends conventional reinforcement learning from human feedback (RLHF) and verifiable rewards (RLVR) by replacing scalar preference signals with...

<https://arxiv.org/abs/2603.15646>

38 CONF

Dynamic Pondering Sparsity-aware Mixture-of-Experts Transformer for Event Stream based Visual Object Tracking

ArXiv CS.AI 50%

arXiv:2605.06112v1 Announce Type: cross Abstract: Despite significant progress, RGB-based trackers remain vulnerable to challenging imaging conditions, such as low illumination and fast motion. Event cameras offer a promising alternative by asynchronously capturing pixel-wise...

<https://arxiv.org/abs/2605.06112>

39 CONF

Listwise Policy Optimization: Group-based RLVR as Target-Projection on the LLM Response Simplex

ArXiv CS.AI 50%

arXiv:2605.06139v1 Announce Type: cross Abstract: Reinforcement learning with verifiable rewards (RLVR) has become a standard approach for large language models (LLMs) post-training to incentivize reasoning capacity. Among existing recipes, group-based policy gradient is...

<https://arxiv.org/abs/2605.06139>

40 CONF

SymDrift: One-Shot Generative Modeling under Symmetries

ArXiv CS.AI 50%

arXiv:2605.06140v1 Announce Type: cross Abstract: Generative modeling of physical systems, such as molecules, requires learning distributions that are invariant under global symmetries, such as rotations in three-dimensional space. Equivariant diffusion and flow matching models...

<https://arxiv.org/abs/2605.06140>

41 CONF

AI-Generated Images: What Humans and Machines See When They Look at the Same Image

ArXiv CS.AI 50%

arXiv:2605.06143v1 Announce Type: cross Abstract: The misuse of generative AI in online disinformation campaigns highlights the urgent need for transparent and explainable detection systems. In this work, we investigate how detectors for AI-generated images can be more effective...

<https://arxiv.org/abs/2605.06143>

42 CONF

HNC: Leveraging Hard Negative Captions towards Models with Fine-Grained Visual-Linguistic Comprehension Capabilities

ArXiv CS.AI 50%

arXiv:2605.06157v1 Announce Type: cross Abstract: Image-Text-Matching (ITM) is one of the defacto methods of learning generalized representations from a large corpus in Vision and Language (VL). However, due to the weak association between the web-collected image-text pairs,...

<https://arxiv.org/abs/2605.06157>

43 CONF

Memory Inception: Latent-Space KV Cache Manipulation for Steering LLMs

ArXiv CS.AI 50%

arXiv:2605.06225v1 Announce Type: cross Abstract: Steering large language models (LLMs) is usually done by either instruction prompting or activation steering. Prompting often gives strong control, but caches guidance tokens at every layer and can clutter long interactions;...

<https://arxiv.org/abs/2605.06225>

44 CONF

Band Together: Untargeted Adversarial Training with Multimodal Coordination against Evasion-based Promotion Attacks

ArXiv CS.AI 50%

arXiv:2605.06238v1 Announce Type: cross Abstract: Multimodal recommender systems exploit visual and textual signals to alleviate data sparsity, but this also makes them more vulnerable to evasion-based promotion attacks. Existing defenses are largely limited to single-modal...

<https://arxiv.org/abs/2605.06238>

45 CONF

Cumulative-Goodness Free-Riding in Forward-Forward Networks: Real, Repairable, but Not Accuracy-Dominant

ArXiv CS.AI 50%

arXiv:2605.06240v1 Announce Type: cross Abstract: Forward-Forward (FF) training allows each layer to learn from a local goodness criterion. In cumulative-goodness variants, however, later layers can inherit a task that earlier layers have already partially separated. We...

<https://arxiv.org/abs/2605.06240>

46 CONF

Inference-Time Refinement Closes the Synthetic-Real Gap in Tabular Diffusion

ArXiv CS.AI 50%

arXiv:2605.06261v1 Announce Type: cross Abstract: Diffusion-based generators set the current state of the art for synthetic tabular data. These methods approach but rarely exceed real-data utility, and closing this synthetic-real gap has so far been pursued exclusively at...

<https://arxiv.org/abs/2605.06261>

47 CONF

Correct Code, Vulnerable Dependencies: A Large Scale Measurement Study of LLM-Specified Library Versions

ArXiv CS.AI 50%

arXiv:2605.06279v1 Announce Type: cross Abstract: Large language models (LLMs) are now largely involved in software development workflows, and the code they generate routinely includes third-party library (TPL) imports annotated with specific version identifiers. These version...

<https://arxiv.org/abs/2605.06279>

48 CONF

Measuring Evaluation-Context Divergence in Open-Weight LLMs: A Paired-Prompt Protocol with Pilot Evidence of Alignment-Pipeline-Specific Heterogeneity

ArXiv CS.AI 50%

arXiv:2605.06327v1 Announce Type: cross Abstract: Safety benchmarks are routinely treated as evidence about how a language model will behave once deployed, but this inference is fragile if behavior depends on whether a prompt looks like an evaluation. We define...

<https://arxiv.org/abs/2605.06327>

49 CONF

Fine-Tuning Small Language Models for Solution-Oriented Windows Event Log Analysis

ArXiv CS.AI 50%

arXiv:2605.06330v1 Announce Type: cross Abstract: Large language models (LLMs) have shown promise for event log analysis, but their high computational requirements, reliance on cloud infrastructure, and security concerns limit practical deployment. In addition, most existing...

<https://arxiv.org/abs/2605.06330>

50 CONF

TinyBayes: Closed-Form Bayesian Inference via Jacobi Prior for Real-Time Image Classification on Edge Devices

ArXiv CS.AI 50%

arXiv:2605.06333v1 Announce Type: cross Abstract: Cocoa (Theobroma cacao) is a critical cash crop for millions of smallholder farmers in West Africa, where Cocoa Swollen Shoot Virus Disease (CSSVD) and anthracnose cause devastating yield losses. Automated disease detection from...

<https://arxiv.org/abs/2605.06333>

51 CONF

Human-AI Co-Evolution and Epistemic Collapse: A Dynamical Systems Perspective

ArXiv CS.AI 50%

arXiv:2605.06347v1 Announce Type: cross Abstract: Large language models (LLMs) are reshaping how knowledge is produced, with increasing reliance on AI systems for generation, summarization, and reasoning. While prior work has studied cognitive offloading in humans and model...

<https://arxiv.org/abs/2605.06347>

52 CONF

Is Escalation Worth It? A Decision-Theoretic Characterization of LLM Cascades

ArXiv CS.AI 50%

arXiv:2605.06350v1 Announce Type: cross Abstract: Model cascades, in which a cheap LLM defers to an expensive one on low-confidence queries, are widely used to navigate the cost-quality tradeoff at deployment. Existing approaches largely treat the deferral threshold as an...

<https://arxiv.org/abs/2605.06350>

53 CONF

eXplaining to Learn (eX2L): Regularization Using Contrastive Visual Explanation Pairs for Distribution Shifts

ArXiv CS.AI 50%

arXiv:2605.06358v1 Announce Type: cross Abstract: Despite extensive research into mitigating distribution shifts, many existing algorithms yield inconsistent performance, often failing to outperform baseline Empirical Risk Minimization (ERM) across diverse scenarios....

<https://arxiv.org/abs/2605.06358>

54 CONF

Continuous-Time Distribution Matching for Few-Step Diffusion Distillation

ArXiv CS.AI 50%

arXiv:2605.06376v1 Announce Type: cross Abstract: Step distillation has become a leading technique for accelerating diffusion models, among which Distribution Matching Distillation (DMD) and Consistency Distillation are two representative paradigms. While consistency methods...

<https://arxiv.org/abs/2605.06376>

55 CONF

Consistent Geometric Deep Learning via Hilbert Bundles and Cellular Sheaves

ArXiv CS.AI 50%

arXiv:2605.06395v1 Announce Type: cross Abstract: Modern deep learning architectures increasingly contend with sophisticated signals that are natively infinite-dimensional, such as time series, probability distributions, or operators, and are defined over irregular domains. Yet,...

<https://arxiv.org/abs/2605.06395>

56 CONF

COVID-19 Infodemic. Understanding content features in detecting fake news using a machine learning approach

ArXiv CS.AI 50%

arXiv:2605.06435v1 Announce Type: cross Abstract: The use of content features, particularly textual and linguistic for fake news detection is under-researched, despite empirical evidence showing the features could contribute to differentiating real and fake news. To this end,...

<https://arxiv.org/abs/2605.06435>

57 CONF

3D MRI Image Pretraining via Controllable 2D Slice Navigation Task

ArXiv CS.AI 50%

arXiv:2605.06487v1 Announce Type: cross Abstract: Self-supervised pretraining has become the mainstream approach for learning MRI representations from unlabeled scans. However, most existing objectives still treat each scan primarily as static aggregations of slices, patches or...

<https://arxiv.org/abs/2605.06487>

58 CONF

Towards Metric-Faithful Neural Graph Matching

ArXiv CS.AI 50%

arXiv:2605.06588v1 Announce Type: cross Abstract: Graph Edit Distance (GED) is a fundamental, albeit NP-hard, metric for structural graph similarity. Recent neural graph matching architectures approximate GED by first encoding graphs with a Graph Neural Network (GNN) and then...

<https://arxiv.org/abs/2605.06588>

59 CONF

DARK: Diagonal-Anchored Repulsive Knowledge Distillation for Vision-Language Models under Extreme Compression

ArXiv CS.AI 50%

arXiv:2603.05421v3 Announce Type: replace-cross Abstract: Compressing vision-language models for on-device deployment is increasingly important in clinical settings, but knowledge distillation (KD) degrades sharply when the teacher-student capacity gap spans an order of...

<https://arxiv.org/abs/2603.05421>

60 CONF

AI CFD Scientist: Toward Open-Ended Computational Fluid Dynamics Discovery with Physics-Aware AI Agents

ArXiv CS.AI 50%

arXiv:2605.06607v1 Announce Type: cross Abstract: Recent LLM-based agents have closed substantial portions of the scientific discovery loop in software-only machine-learning research, in chemistry, and in biology. Extending the same loop to high-fidelity physical simulators is...

<https://arxiv.org/abs/2605.06607>

61 CONF

Are We Making Progress in Multimodal Domain Generalization? A Comprehensive Benchmark Study

ArXiv CS.AI 50%

arXiv:2605.06643v1 Announce Type: cross Abstract: Despite the growing popularity of Multimodal Domain Generalization (MMDG) for enhancing model robustness, it remains unclear whether reported performance gains reflect genuine algorithmic progress or are artifacts of inconsistent...

<https://arxiv.org/abs/2605.06643>

62 CONF

When No Benchmark Exists: Validating Comparative LLM Safety Scoring Without Ground-Truth Labels

ArXiv CS.AI 50%

arXiv:2605.06652v1 Announce Type: cross Abstract: Many deployments must compare candidate language models for safety before a labeled benchmark exists for the relevant language, sector, or regulatory regime. We formalize this setting as benchmarkless comparative safety scoring...

<https://arxiv.org/abs/2605.06652>

63 CONF

Optimizer-Model Consistency: Full Finetuning with the Same Optimizer as Pretraining Forgets Less

ArXiv CS.AI 50%

arXiv:2605.06654v1 Announce Type: cross Abstract: Optimizers play an important role in both pretraining and finetuning stages when training large language models (LLMs). In this paper, we present an observation that full finetuning with the same optimizer as in pretraining...

<https://arxiv.org/abs/2605.06654>

64 CONF

BAMI: Training-Free Bias Mitigation in GUI Grounding

ArXiv CS.AI 50%

arXiv:2605.06664v1 Announce Type: cross Abstract: GUI grounding is a critical capability for enabling GUI agents to execute tasks such as clicking and dragging. However, in complex scenarios like the ScreenSpot-Pro benchmark, existing models often suffer from suboptimal...

<https://arxiv.org/abs/2605.06664>

65 CONF

Flexible Agent Alignment with Goal Inference from Open-Ended Dialog

ArXiv CS.AI 50%

arXiv:2508.15119v2 Announce Type: replace Abstract: We introduce Open-Universe Assistance Games (OU-AGs), a formal framework extending assistance games to LLM-based agents. Effective assistance requires reasoning over human preferences that are unbounded, underspecified, and...

<https://arxiv.org/abs/2508.15119>

66 CONF

CompassLLM: A Multi-Agent Approach toward Geo-Spatial Reasoning for Popular Path Query

ArXiv CS.AI 50%

arXiv:2510.07516v2 Announce Type: replace Abstract: The popular path query - identifying the most frequented routes between locations from historical trajectory data - has important applications in urban planning, navigation optimization, and travel recommendations. While...

<https://arxiv.org/abs/2510.07516>

67 CONF

Self-Consistency Is Losing Its Edge: Diminishing Returns and Rising Costs in Modern LLMs

ArXiv CS.AI 50%

arXiv:2511.00751v2 Announce Type: replace Abstract: Self-consistency -- sampling multiple reasoning paths and selecting the most frequent answer -- was designed for an era when language models made frequent, unpredictable errors. This study argues that the technique has become...

<https://arxiv.org/abs/2511.00751>

68 CONF

ProAgent: Harnessing On-Demand Sensory Contexts for Proactive LLM Agent Systems in the Wild

ArXiv CS.AI 50%

arXiv:2512.06721v2 Announce Type: replace Abstract: Recent studies have begun to explore proactive large language model (LLM) agents that provide unobtrusive assistance by automatically leveraging contextual information, such as in code editing and in-app suggestions. However,...

<https://arxiv.org/abs/2512.06721>

69 CONF

AI Agents Alone Are Not (Yet) Sufficient for Social Simulation

ArXiv CS.AI 50%

arXiv:2603.00113v2 Announce Type: replace-cross Abstract: Recent advances in large language models (LLMs) have spurred growing interest in using LLM-integrated agents for social simulation, often under the implicit assumption that realistic population dynamics will emerge once...

<https://arxiv.org/abs/2603.00113>

70 CONF

Latent Generative Solvers for Generalizable Long-Term Physics Simulation

ArXiv CS.AI 50%

arXiv:2602.11229v2 Announce Type: replace Abstract: Reliable physics simulation demands two capabilities that today's neural PDE solvers do not deliver together: generalization across heterogeneous PDE families, and stability under long autoregressive rollouts. Deterministic...

<https://arxiv.org/abs/2602.11229>

71 CONF

Beyond Factual Correctness: Mitigating Preference-Inconsistent Explanations in Explainable Recommendation

ArXiv CS.AI 50%

arXiv:2603.03080v2 Announce Type: replace Abstract: LLM-based explainable recommenders can produce fluent explanations that are factually correct, yet still justify items using attributes that conflict with a user's historical preferences. Such preference-inconsistent...

<https://arxiv.org/abs/2603.03080>

72 CONF

Supervising Ralph Wiggum: Exploring a Metacognitive Co-Regulation Agentic AI Loop for Engineering Design

ArXiv CS.AI 50%

arXiv:2603.24768v2 Announce Type: replace Abstract: The engineering design research community has studied agentic AI systems that use Large Language Model (LLM) agents to automate the engineering design process. However, these systems are prone to some of the same pathologies...

<https://arxiv.org/abs/2603.24768>

73 CONF

Benchmarking PNW Model for MedMNIST to 100% Accuracy

ArXiv CS.AI 50%

arXiv:2604.18916v4 Announce Type: replace Abstract: In this paper, we introduce a new concept called Artificial Special Intelligence by which Machine Learning models for the classification problem can be trained error-free, thus acquiring the capability of not making repeated...

<https://arxiv.org/abs/2604.18916>

74 CONF

ReMAP: Neural Reparameterization for Scalable MAP Inference in Arbitrary-Order Markov Random Fields

ArXiv CS.AI 50%

arXiv:2411.18954v4 Announce Type: replace-cross Abstract: Scalable high-quality MAP inference in arbitrary-order Markov Random Fields (MRFs) remains challenging. Approximate message-passing methods are often efficient but can degrade on dense or high-order instances, while exact...

<https://arxiv.org/abs/2411.18954>

75 CONF

LicenseGPT: A Fine-tuned Foundation Model for Publicly Available Dataset License Compliance

ArXiv CS.AI 50%

arXiv:2501.00106v2 Announce Type: replace-cross Abstract: Dataset license compliance is a critical yet complex aspect of developing commercial AI products, particularly with the increasing use of publicly available datasets. Ambiguities in dataset licenses pose significant legal...

<https://arxiv.org/abs/2501.00106>

76 CONF

Toward Practical Equilibrium Propagation: Brain-inspired Recurrent Neural Network with Feedback Regulation and Residual Connections

ArXiv CS.AI 50%

arXiv:2508.11659v2 Announce Type: replace-cross Abstract: Brain-like intelligent systems need brain-like learning methods. Equilibrium Propagation (EP) is a biologically plausible learning framework with strong potential for brain-inspired computing hardware. However, existing...

<https://arxiv.org/abs/2508.11659>

77 CONF

REMAP: Regularized Matching and Partial Alignment of Video Embeddings

ArXiv CS.AI 50%

arXiv:2509.24382v2 Announce Type: replace-cross Abstract: Real-world instructional videos are long, noisy, and often contain extended background segments, repeated actions, and execution variability that do not correspond to meaningful procedural steps. We propose **REMAP**, an...

<https://arxiv.org/abs/2509.24382>

78 CONF

Beyond Value Elicitation: Towards Moral Profiles in Early Requirements Engineering via Role-Playing Games and Anthropologist LLMs

ArXiv CS.AI 50%

arXiv:2510.01189v2 Announce Type: replace-cross Abstract: This study presents a proof of concept for eliciting and representing the moral profiles of digital system users in Requirements Engineering (RE) by combining immersive role-playing games (RPGs) with large language model...

<https://arxiv.org/abs/2510.01189>

79 CONF

RobustSora: De-Watermarked Benchmark for Robust AI-Generated Video Detection

ArXiv CS.AI 50%

arXiv:2512.10248v2 Announce Type: replace-cross Abstract: The proliferation of AI-generated video models poses new challenges to information integrity and digital trust. A key confound, however, remains unaddressed: commercial generators embed visible overlay watermarks for...

<https://arxiv.org/abs/2512.10248>

80 CONF

Interpretability-Guided Bi-objective Optimization: Aligning Accuracy and Explainability

ArXiv CS.AI 50%

arXiv:2601.00655v3 Announce Type: replace-cross Abstract: This paper introduces Interpretability-Guided Bi-objective Optimization (IGBO), a framework that trains interpretable models by incorporating structured domain knowledge via a bi-objective formulation. IGBO encodes...

<https://arxiv.org/abs/2601.00655>

81 CONF

Beyond Fixed Psychological Personas: State Beats Trait, but Language Models are State-Blind

ArXiv CS.AI 50%

arXiv:2601.15395v2 Announce Type: replace-cross Abstract: User interactions with language models vary due to static properties of the user (trait) and the specific context of the interaction (state). However, existing persona datasets (like PersonaChat, PANDORA etc.) capture...

<https://arxiv.org/abs/2601.15395>

82 CONF

AI Cap-and-Trade: Efficiency Incentives for Accessibility and Sustainability

ArXiv CS.AI 50%

arXiv:2601.19886v2 Announce Type: replace-cross Abstract: The race for artificial intelligence (AI) dominance often prioritizes scale over efficiency. Hyper-scaling is the common industry approach: larger models, more data, and as many computational resources as possible. Using...

<https://arxiv.org/abs/2601.19886>

83 CONF

Theoretically Optimal Attention/FFN Ratios in Disaggregated LLM Serving

ArXiv CS.AI 50%

arXiv:2601.21351v2 Announce Type: replace-cross Abstract: Attention-FFN disaggregation (AFD) is an emerging architecture for LLM decoding that separates state-heavy, KV-cache-dominated Attention computation from stateless, compute-intensive FFN computation, connected by per-step...

<https://arxiv.org/abs/2601.21351>

84 CONF

Conversation for Non-verifiable Learning: Self-Evolving LLMs through Meta-Evaluation

ArXiv CS.AI 50%

arXiv:2601.21464v2 Announce Type: replace-cross Abstract: Training large language models (LLMs) for non-verifiable tasks, such as creative writing, dialogue, and ethical reasoning, remains challenging due to the absence of ground-truth labels. While LLM-as-Judge approaches offer...

<https://arxiv.org/abs/2601.21464>

85 CONF

The Illusion of Forgetting: Attack Unlearned Diffusion via Initial Latent Variable Optimization

ArXiv CS.AI 50%

arXiv:2602.00175v2 Announce Type: replace-cross Abstract: Text-to-image diffusion models (DMs) are frequently abused to produce harmful or copyrighted content, violating public interests. Concept erasure (unlearning) is a promising paradigm to alleviate this issue. However,...

<https://arxiv.org/abs/2602.00175>

86 CONF

SMI: Statistical Membership Inference for Reliable Unlearned Model Auditing

ArXiv CS.AI 50%

arXiv:2602.01150v2 Announce Type: replace-cross Abstract: Machine unlearning (MU) is essential for enforcing the right to be forgotten in machine learning systems. A key challenge of MU is how to reliably audit whether a model has truly forgotten specified training data....

<https://arxiv.org/abs/2602.01150>

87 CONF

PixelGen: Improving Pixel Diffusion with Perceptual Supervision

ArXiv CS.AI 50%

arXiv:2602.02493v2 Announce Type: replace-cross Abstract: Pixel diffusion generates images directly in pixel space, avoiding the VAE artifacts and representational bottlenecks of two-stage latent diffusion. Recent JiT further simplifies pixel diffusion with x-prediction, where...

<https://arxiv.org/abs/2602.02493>

88 CONF

Parity, Sensitivity, and Transformers

ArXiv CS.AI 50%

arXiv:2602.05896v2 Announce Type: replace-cross Abstract: Understanding what neural architectures can and cannot compute is a central challenge in the theory of AI. One of the fundamental problems in this context is the PARITY task, which asks whether the number of 1s in a...

<https://arxiv.org/abs/2602.05896>

89 CONF

AceGRPO: Adaptive Curriculum Enhanced Group Relative Policy Optimization for Autonomous Machine Learning Engineering

ArXiv CS.AI 50%

arXiv:2602.07906v5 Announce Type: replace-cross Abstract: Autonomous Machine Learning Engineering (MLE) requires agents to perform sustained, iterative optimization over long horizons. While recent LLM-based agents show promise, current prompt-based agents for MLE suffer from...

<https://arxiv.org/abs/2602.07906>

90 CONF

Multimodal Fact-Level Attribution for Verifiable Reasoning

ArXiv CS.AI 50%

arXiv:2602.11509v2 Announce Type: replace-cross Abstract: Multimodal large language models (MLLMs) are increasingly used for real-world tasks involving multi-step reasoning and long-form generation, where reliability requires grounding model outputs in heterogeneous input...

<https://arxiv.org/abs/2602.11509>

91 CONF

Molecular Design beyond Training Data with Novel Extended Objective Functionals of Generative AI Models Driven by Quantum Annealing Computer

ArXiv CS.AI 50%

arXiv:2602.15451v2 Announce Type: replace-cross Abstract: Deep generative modeling to stochastically design small molecules is an emerging technology for accelerating drug discovery and development. However, one major issue in molecular generative models is their lower frequency...

<https://arxiv.org/abs/2602.15451>

92 CONF

Perceptive Humanoid Parkour: Chaining Dynamic Human Skills via Motion Matching

ArXiv CS.AI 50%

arXiv:2602.15827v2 Announce Type: replace-cross Abstract: While recent advances in humanoid locomotion have achieved stable walking on varied terrains, capturing the agility and adaptivity of highly dynamic human motions remains an open challenge. In particular, agile parkour in...

<https://arxiv.org/abs/2602.15827>

93 CONF

Decentralized Attention Fails Centralized Signals: Rethinking Transformers for Medical Time Series

ArXiv CS.AI 50%

arXiv:2602.18473v2 Announce Type: replace-cross Abstract: Accurate analysis of medical time series (MedTS) data, such as electroencephalography (EEG) and electrocardiography (ECG), plays a pivotal role in healthcare applications, including the diagnosis of brain and heart...

<https://arxiv.org/abs/2602.18473>

94 CONF

Path Dependence under Adaptive AI Delegation

ArXiv CS.AI 50%

arXiv:2603.02950v2 Announce Type: replace-cross Abstract: Repeated AI assistance can improve immediate task performance while reducing the skill available for future independent work. We develop a mathematical framework for this long-run tradeoff. The model tracks two state...

<https://arxiv.org/abs/2603.02950>

95 CONF

From Documents to Spans: Scalable Supervision for Evidence-Based ICD Coding with LLMs

ArXiv CS.AI 50%

arXiv:2603.15270v2 Announce Type: replace-cross Abstract: International Classification of Diseases (ICD) coding assigns diagnosis codes to clinical documents and is essential for healthcare billing and clinical analysis. Reliable coding requires that each predicted code be...

<https://arxiv.org/abs/2603.15270>

96 CONF

Structural Sensitivity in Compressed Transformers: Relative Error Propagation and Layer Removal

ArXiv CS.AI 50%

arXiv:2603.20991v2 Announce Type: replace-cross Abstract: Compressing transformer weights makes large language models cheaper to deploy. But each layer's compression introduces an error. These errors accumulate as the signal passes through later layers, and how they accumulate...

<https://arxiv.org/abs/2603.20991>

97 CONF

Spectral Edge Dynamics: An Analytical-Empirical Study of Phase Transitions in Neural Network Training

ArXiv CS.AI 50%

arXiv:2603.28964v3 Announce Type: replace-cross Abstract: We develop the spectral edge analysis: phase transitions in neural network training -- grokking, capability gains, loss plateaus -- are controlled by the spectral gap of the rolling-window Gram matrix of parameter...

<https://arxiv.org/abs/2603.28964>

98 CONF

StableTTA: Improving Vision Model Performance by Training-free Test-Time Adaptation Methods

ArXiv CS.AI 50%

arXiv:2604.04552v3 Announce Type: replace-cross Abstract: Ensemble methods improve predictive performance but often incur high memory and computational costs. We identify an aggregation instability induced by nonlinear projection and voting operations. To address both efficiency...

<https://arxiv.org/abs/2604.04552>

99 CONF

Reward Score Matching: Unifying Reward-based Fine-tuning for Flow and Diffusion Models

ArXiv CS.AI 50%

arXiv:2604.17415v2 Announce Type: replace-cross Abstract: Reward-based fine-tuning steers a pretrained diffusion or flow-based generative model toward higher-reward samples while remaining close to the pretrained model. Although existing methods are derived from different...

<https://arxiv.org/abs/2604.17415>

00 CONF

How Fast Should a Model Commit to Supervision? Training Reasoning Models on the Tsallis Loss Continuum

ArXiv CS.AI 50%

arXiv:2604.25907v2 Announce Type: replace-cross Abstract: SFT-then-RLVR is widely used for post-training reasoning models, but why this specific ordering, and why RLVR-only stalls at cold start, have lacked a unifying theoretical account. We provide that account under a unified...

<https://arxiv.org/abs/2604.25907>

01 CONF

InvEvolve: Evolving White-Box Inventory Policies via Large Language Models with Performance Guarantees

ArXiv CS.AI 50%

arXiv:2605.00369v2 Announce Type: replace-cross Abstract: We study how large language models can be used to evolve inventory policies in online, non-stationary environments. Our work is motivated by recent advances in LLM-based evolutionary search, such as AlphaEvolve, which...

<https://arxiv.org/abs/2605.00369>

02 CONF

Predictive and Prescriptive AI toward Optimizing Wildfire Suppression

ArXiv CS.AI 50%

arXiv:2605.04510v2 Announce Type: replace-cross Abstract: Intense wildfire seasons require critical prioritization decisions to allocate scarce suppression resources over a dispersed geographical area. This paper develops a predictive and prescriptive approach to jointly...

<https://arxiv.org/abs/2605.04510>

03 CONF

Continual Knowledge Updating in LLM Systems: Learning Through Multi-Timescale Memory Dynamics

ArXiv CS.AI 50%

arXiv:2605.05097v2 Announce Type: replace-cross Abstract: LLMs are trained once, then deployed into a world that never stops changing. External memory compensates for this, but most systems manage it explicitly rather than letting it adapt on its own. Biological memory works...

<https://arxiv.org/abs/2605.05097>